All PID gains: Success is in the green region on every axis

Failure 1

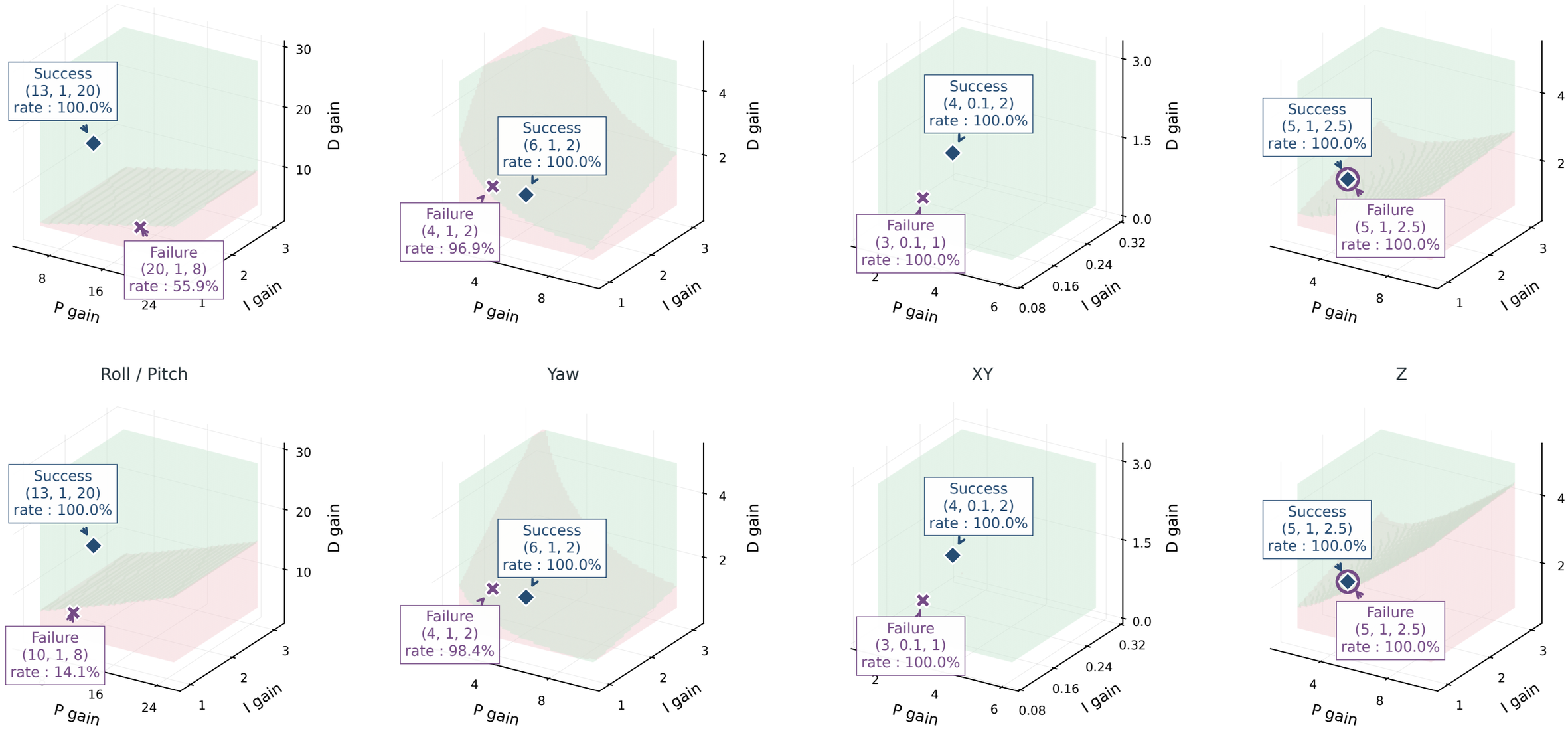
Failure 2

Roll / Pitch

Yaw

XY

Z



3D volumes are cut at each shared reference I to show both points. Purple marks identify the failure-flight gains; Z gains coincide.

Each case uses its own posterior. Other controller gains are fixed at the successful-flight values.